

Last Time

Confidence Intervals

When σ^2 is Unknown

Other Confidence Intervals: Variance

Hypothesis Testing: Single Means

Research vs. Statistical Questions

The Null Hypothesis

The p -value

The Z-test

Test of Proportions

Today

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Test of Proportions

Z- and t -scores

- ▶ Both Z and t express a variable's **distance from its mean** in units of **standard deviations**.
- ▶ For some variable V (X , $\bar{X}$, etc.),

$$z_V = \frac{V - \mu_V}{\sigma_V}$$

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- ▶ The farther a value of a measurement V is from the mean of its distribution (μ_V), the smaller the probability of observing measurements as extreme as that value (Chebyshev's Inequality).

Tables

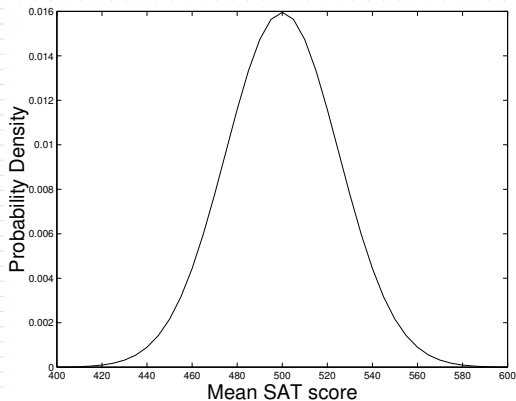
Download these tables in case of dystopian future:

`pnorm()` Normal table

`pt(...,  $\nu$ )` T table

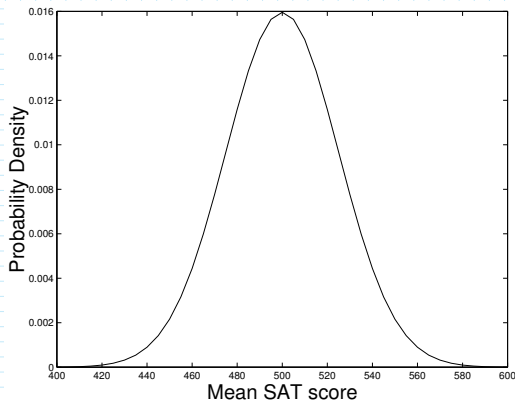
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What is the probability of obtaining a sample mean $\bar{X}$ *as extreme or more extreme* than the one we computed for our sample ($\bar{x}$) given that this is what the sampling distribution of the mean must be?

Hypothesis Testing

- ▶ Hypothesis testing is a procedure for deciding if an hypothesis about a population parameter (like μ) is true given the value of the statistic that we compute from a sample.
- ▶ The result of an hypothesis test is a *probablistic* answer to our question: we can compute probabilities associated with our statistical value if the hypothesis is true.
- ▶ Statistical questions are not research questions. A “significant” hypothesis test does not imply an important or interesting research question.

Research vs. statistical questions

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- ▶ Research question example: on standardized exams, should you answer with the first response that comes to mind, or is it okay to go back and change your answers?
- ▶ Example: Consider a standardized test of academic achievement administered to 5th graders. The national average on this test is $\mu = 200$ and the national standard deviation is $\sigma = 48$.
- ▶ A class of $N = 64$ 5th graders at Indianola Elementary is instructed to give the first answers that come to mind. The average performance was $\bar{X} = 220$.
- ▶ Statistical question: is $\bar{X} = 220$ significantly greater than 200?

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- ▶ Alternatively, the treatment had the desired effect (improved performance).

$$H_1 : \quad \mu > 200$$

One- vs. Two-Tailed Tests

- ▶ The null hypothesis H_0 always contains the uninteresting result (an “=” sign):

$$H_0 : \mu = \mu_0$$

- ▶ Two-tailed test:

$$H_1 : \mu \neq \mu_0$$

- ▶ One-tailed test:

$$H_1 : \mu > \mu_0 \text{ OR } H_1 : \mu < \mu_0$$

- ▶ The scientific question determines the direction of the alternative.
- ▶ Two-tailed tests are rarely interesting.

The p -value

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- ▶ BUT what does “highly unlikely” mean?
 - ▶ We must select a criterion probability against which to evaluate $Pr(\bar{X} \geq \bar{x} \mid \mu = \mu_0)$.
 - ▶ This criterion is called the α -level and is traditionally set to 0.05 (0.005).
 - ▶ The α -level is the probability of making a mistake (Type I error).

The α -Level

- ▶ We need to set rules about when H_0 will be *rejected* so that our probability of making a mistake is
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- ▶ Significance level = $\alpha = P(\text{Type I error} \mid H_0 \text{ true})$

Errors in Hypothesis Testing

How many ways can we make a mistake in our statistical conclusion?

Errors in Hypothesis Testing

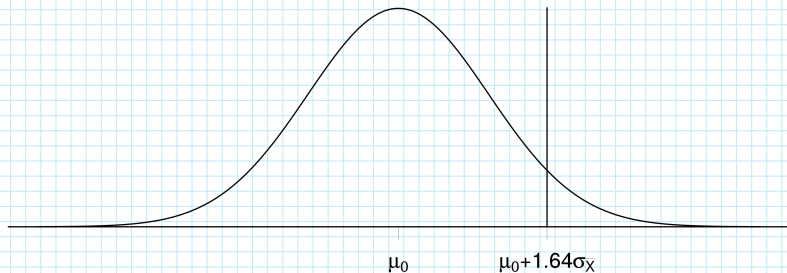
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Two ways:

State of the world	Decision	
	Fail to reject H_0	Reject H_0
H_0 True	Correct	Type I Error
H_0 False	Type II Error	Correct

Incorrectly Rejecting H_0

We expect to see extreme, unlikely means about $100\alpha\%$ of the time.



Values of $\bar{X}$ if H_0 is true

If we reject H_0 incorrectly, we have committed a Type I error.

The Z-test

- ▶ Testing hypotheses about single means *when σ is known*.
- ▶ Achievement Test: For $N = 64$, we observe $\bar{X} = 220$. The national average of these scores is $\mu = 200$ and the standard deviation is $\sigma = 48$.

Three Ways to Do A Z-test

1. **Compare the p -value of $\bar{X}$ to α -level.**
2. Compare the z -score of $\bar{X}$ to the z -score associated with the desired α -level.
3. Compare $\bar{X}$ to the value of the mean that has the z -score associated with the desired α -level.

Use Z-score method

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- ▶ If this probability is smaller than the desired α -level, reject H_0 .
- ▶ OR: Determine the *critical values* of the test statistic as determined by the α -level.
- ▶ Compare the test statistic to the critical values. If the test statistic is
 - ▶ more extreme than the critical values, reject H_0 .
 - ▶ not as extreme as the critical values, fail to reject H_0 .

Test of Proportions

Let $\{X_1, X_2, \dots, X_N\}$ be a sequence of (0,1) Bernoulli(p) observations, where p is the population probability of success. Then $\bar{X} = \hat{p}$, the proportion of “successes” in the sample and an estimate of p .

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We know that

$$\begin{aligned} E(X) = \mu &= p \text{ and} \\ \text{Var}(X) = \sigma^2 &= p(1 - p). \end{aligned}$$

So $E(\bar{X}) = E(X) = p$ and $\text{Var}(\bar{X}) = \sigma^2/N = p(1 - p)/N$.

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Central Limit Theorem: for large enough N , $\bar{X} = \hat{p} \sim \mathcal{N}(p, p(1 - p)/N)$.

This is one of those rare circumstances where we know σ^2 : the null hypothesis $H_0 : p = p_0$ determines both the mean and the variance of the sampling distribution of $\bar{X} = \hat{p}$.

Birth Order Effects

$$H_0 : p \leq 0.5$$

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Set $\alpha = 0.05$.

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$$\bar{X} = 10/15 = 0.6667$$

$$\sigma_{\bar{X}}^2 = 0.5(1 - 0.5)/15 = 0.0167$$

$$Z_{\bar{X}} = \frac{0.6667 - 0.50}{\sqrt{0.0167}} = 1.2910$$

$$Pr(Z \geq 1.2910) = 0.0984$$

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Birth Order Effects

```
prop.test(10,15,p=.5,alternative="greater")
```

1-sample proportions test with continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.0667, df = 1, p-value = 0.1508

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4234848 1.0000000

sample estimates:

p

0.6666667

Remember the Correction For Continuity

Compare:

```
prop.test(10,15,p=.5,alternative="greater",correct=FALSE)
```

1-sample proportions test without continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.6667, df = 1, p-value = 0.09835

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4551721 1.0000000

sample estimates:

p

0.6666667

Yates Correction

We computed $Pr(\hat{p} > 0.6667 \mid p_0 = .5) = Pr(Z > 1.2910) = 0.0984$.

But $\hat{p} = 0.6667$ was computed as $\frac{10}{15}$. Because $\bar{X} = \hat{p}$ is discrete,

$$Pr(\hat{p} \geq 10/15 \mid p_0 = 0.5) = Pr(\hat{p} > 9.5/15 \mid p_0 = 0.5).$$

The *real lower limit* of 10 is 9.5, which is where we need to consider the starting point to compute the area under the (continuous) normal density function.

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Recompute:

$$\hat{p} = 9.5/15 = 0.6333, \text{ so}$$

$$Z_{\hat{p}} = \frac{0.6333 - 0.5}{\sqrt{0.0167}} = 1.0318$$

$$Pr(Z \geq 1.0318) = 0.1508$$

Exact Binomial Test

The p -value with the Yates correction is 0.1508.

Compare this to the exact binomial test:

$$Pr(X \geq 10 \mid p = 0.5) = 0.1509$$

```
binom.test(10,15,alternative="greater")
```

```
Exact binomial test
```

```
data: 10 and 15
```

```
number of successes = 10, number of trials = 15, p-value = 0.1509
```

```
alternative hypothesis: true probability of success is greater than 0.5
```

```
95 percent confidence interval:
```

```
0.4225563 1.0000000
```

```
sample estimates:
```

```
probability of success
```

```
0.6666667
```